

HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026 • CODE & PROTOTYPE SUBMISSION

OS4All — Digital Twin for Preventive Health

"Connecting longitudinal health history with dynamic physiological signals to identify subclinical health drift and project near-term adverse metabolic risk."

 PARTICIPANT

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Solo Participant (Team Size: 1)

 INSTITUTION

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 HEALTHCARE USE CASE

Type 2 Diabetes Mellitus

Near-Term Glucose Spike Horizon

 TECHNOLOGY FOUNDATION

Spring Boot 3 + Flutter

Synthea FHIR + Google GenAI SDK



Reactive Healthcare Model

CLINICAL DELAY

- **Symptom-driven treatment:** Interventions typically occur only after acute symptoms appear or periodic blood tests reveal established clinical complications.
- **Silent progression:** Chronic metabolic conditions (Type 2 Diabetes, hypertension) drift silently over months without causing acute complaints.
- **Missed prevention window:** Without continuous baseline tracking, early reversible physiological departures go undetected.



Fragmented Data Silos

INFORMATION GAP

- **Disconnected records:** Longitudinal EHR records (diagnoses, HbA1c, lab history) live in clinic databases, disconnected from everyday life.
- **Isolated wearables:** Dynamic physiological telemetry (CGM, heart rate, sleep, HRV) lives in consumer smartwatch applications.
- **No unified virtual view:** Clinicians lack a single model uniting chronic patient susceptibility with acute daily lifestyle stressors.



Population vs. Personal Baselines

INDIVIDUAL VARIATION

- **Rigid reference intervals:** Conventional apps compare biometrics against static population thresholds (e.g., resting HR 60–100 bpm).
- **The individual reality:** *Normal for the population is not normal for the individual.* An athletic baseline of 48 bpm drifting to 65 bpm (+35%) indicates major stress, yet is labeled "normal".
- **Need for personalized envelopes:** Personal rolling means and Z-scores are essential to detect true subclinical drift.

The Digital Twin Paradigm Shift:

Instead of disconnected snapshot readings, a healthcare **Digital Twin** maintains a stateful, continuously synchronized virtual patient model that fuses static EHR context with dynamic telemetry to detect early drift and project near-term adverse trajectories.

Prevention • Baselines • Projections

Stream 1: Static / Longitudinal EHR

SYNTHEA FHIR R4

Represents the patient's longitudinal clinical background & underlying vulnerability:

- **Patient Demographics:** Age, biological sex, height, weight, baseline BMI.
- **Confirmed Chronic Diagnoses:** SNOMED-CT codes for Type 2 Diabetes Mellitus (44054006), Essential Hypertension (59621000).
- **Longitudinal Laboratory Panels:** Retrospective LOINC HbA1c (4548-4, baseline 7.6%), fasting blood glucose, lipid panel, serum creatinine.
- **Active Pharmacotherapy:** RxNorm medication records (Metformin hydrochloride, baseline insulin therapy).

Source: Synthetic de-identified patient cohorts generated via Synthea™ FHIR.

Stream 2: Dynamic Telemetry & IoT

5-MIN SIMULATED / IOT

Represents acute physiological responses to meals, stress, sleep, and physical activity:

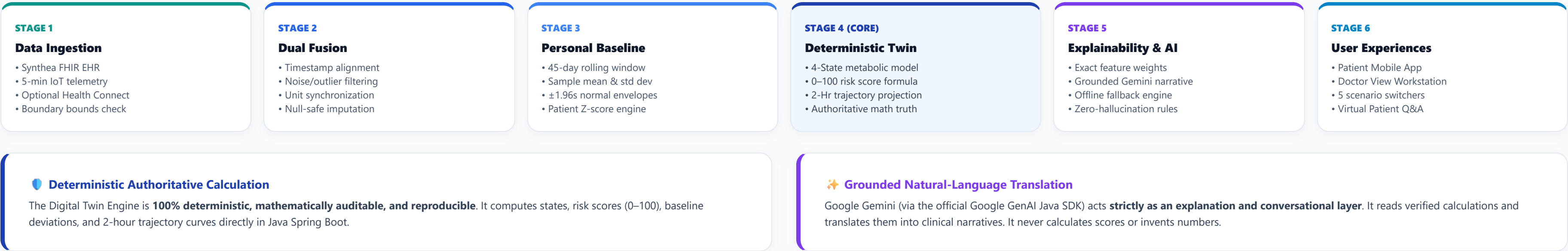
- **Interstitial CGM Glucose:** Continuous 5-minute telemetry packets (valid range 30.0–500.0 mg/dL).
- **Glucose Velocity (dG/dt):** Rate of change in mg/dL/min (+ positive = rising surge).
- **Autonomic Biomarkers:** Heart Rate (bpm) and RMSSD Heart Rate Variability (ms) tracking sympathetic/parasympathetic tone.
- **Sleep Architecture:** Sleep duration, sleep efficiency, and acute cumulative debt.
- **Optional Real Hardware:** Native Android Health Connect read-only sync.

Source: 5-minute simulated telemetry engine with 5 controllable perturbation scenarios.

The Clinical Goal of the Use Case:

Fuse the patient's chronic glycemic profile with acute multi-signal stress (e.g. 4.5h sleep debt + post-meal glucose velocity) to detect **Pre-Symptomatic Drift** and project a **2-Hour Forward Glucose Trajectory**, enabling lifestyle mitigation (such as a 20-minute walk) before acute metabolic excursions peak.

2-Hour Early Horizon



Personal Baseline Engine

45-DAY ROLLING

Replaces generic population intervals with individual statistics:

- **Sample Adequacy:** Requires minimum $N \geq 5$ valid readings to establish baseline.
- **Tukey's Fences Outlier Filtering:** Rejects sensor artifacts outside $[Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR]$.
- **Normal Operating Envelopes:** Calculated with Bessel's unbiased correction:

$$\text{Operating Envelope} = [\bar{x} - 1.96 \cdot s, \bar{x} + 1.96 \cdot s]$$

Patient-Specific Z-Score Normalization:

$$Z = (x_{\text{current}} - \bar{x}_{\text{baseline}}) / s$$

Applied dynamically to CGM glucose, resting HR, HRV, sleep, and steps.

Metabolic State & Risk Engine

DETERMINISTIC

4 Discrete Physiological States:

• **STABLE**
Homeostatic equilibrium; biometrics within personal $\pm 1.96s$ bounds.

• **PRE_SYMPTOMATIC_DRIFT**
Autonomic stress, suppressed HRV, acute cumulative sleep deficit.

• **ELEVATED_RISK**
Rapid glucose velocity & autonomic strain; high spike probability.

• **ACTIVE_ANOMALY**
Severe metabolic excursion exceeding critical physiological thresholds.

Authoritative 0–100 Weighted Risk Scoring:

- Glucose Velocity & Level: **30%**
- Autonomic HRV Suppression: **20%**
- Resting HR Drift: **10%**
- Glycemic CV%: **20%**
- Cumulative Sleep Debt: **15%**
- Chronic BMI Factor: **5%**

Note: The 0–100 score is a deterministic prototype risk index for demonstration, not a clinically calibrated disease probability.



Damped Velocity Decay Formulation

DIFFERENTIAL MODELING

Glucose rate of change rarely stays constant. Endogenous counter-regulation, tissue uptake, and clearance kinetics dampen velocity over time:

$$v_{\text{eff}}(t) = (v_0 + v_{\text{activity}}) \cdot e^{(-\lambda \cdot t)}$$

Integrating effective velocity yields the projected glucose level across the 120-minute forward window:

$$G(t) = G_0 + (v_0 + v_{\text{activity}}) \cdot [(1 - e^{(-\lambda \cdot t)}) / \lambda]$$

- Initial State:** Current 5-min interstitial CGM glucose (G_0) & velocity (v_0 in mg/dL/min).
- Clearance Rate (λ):** $\lambda = 0.010 \text{ min}^{-1}$ for Type 2 Diabetes profiles (prolonged excursion).
- GLUT4 Activity Modifier (v_{act}):** -0.12 mg/dL/min during physical activity (e.g. post-meal walk).



5 Discrete Projection Milestones

TEMPORAL HORIZON

Provides actionable temporal visibility rather than an abstract scalar percentage:

t = 0 min

Current

Observed CGM

+30 min

Early Rise

Initial peak

+60 min

Peak Crest

Max velocity decay

+90 min

Sustained

Metabolic plateau

+120 min

2-Hr Horizon

Return curve

Clinician Value: Allows doctors to evaluate whether an acute glucose surge will exceed 180 mg/dL or self-correct, and models the immediate dampening benefit of walking.

Implemented in MetabolicPredictionEngine.java • Validated across 87+ Spring Boot unit/integration test suites.

Layer 1: Deterministic Twin Engine

AUTHORITATIVE TRUTH

Sole Numerical Source of Truth:

- **Owns All Mathematical Outputs:** Computes physiological state transitions, 0–100 risk score, 2-hour trajectory milestone points, and Z-score deviations.
- **Feature Attribution Table:** Deconstructs exact risk score points (+/-), effect direction (RISK_INCREASING vs. PROTECTIVE), and physiological rationale:

- Glucose Velocity (+1.8 mg/dL/min) → +24 pts (Risk Increasing)
 - Sleep Debt (4.5h vs 7.8h) → +15 pts (Autonomic Strain)
 - Step Count (12,400 steps) → -12 pts (Protective GLUT4)
- **Auditable & Reproducible:** 100% deterministic code; zero black-box variability.

Layer 2: Google Gemini Explanation Layer

GEMINI-3.8-FLASH

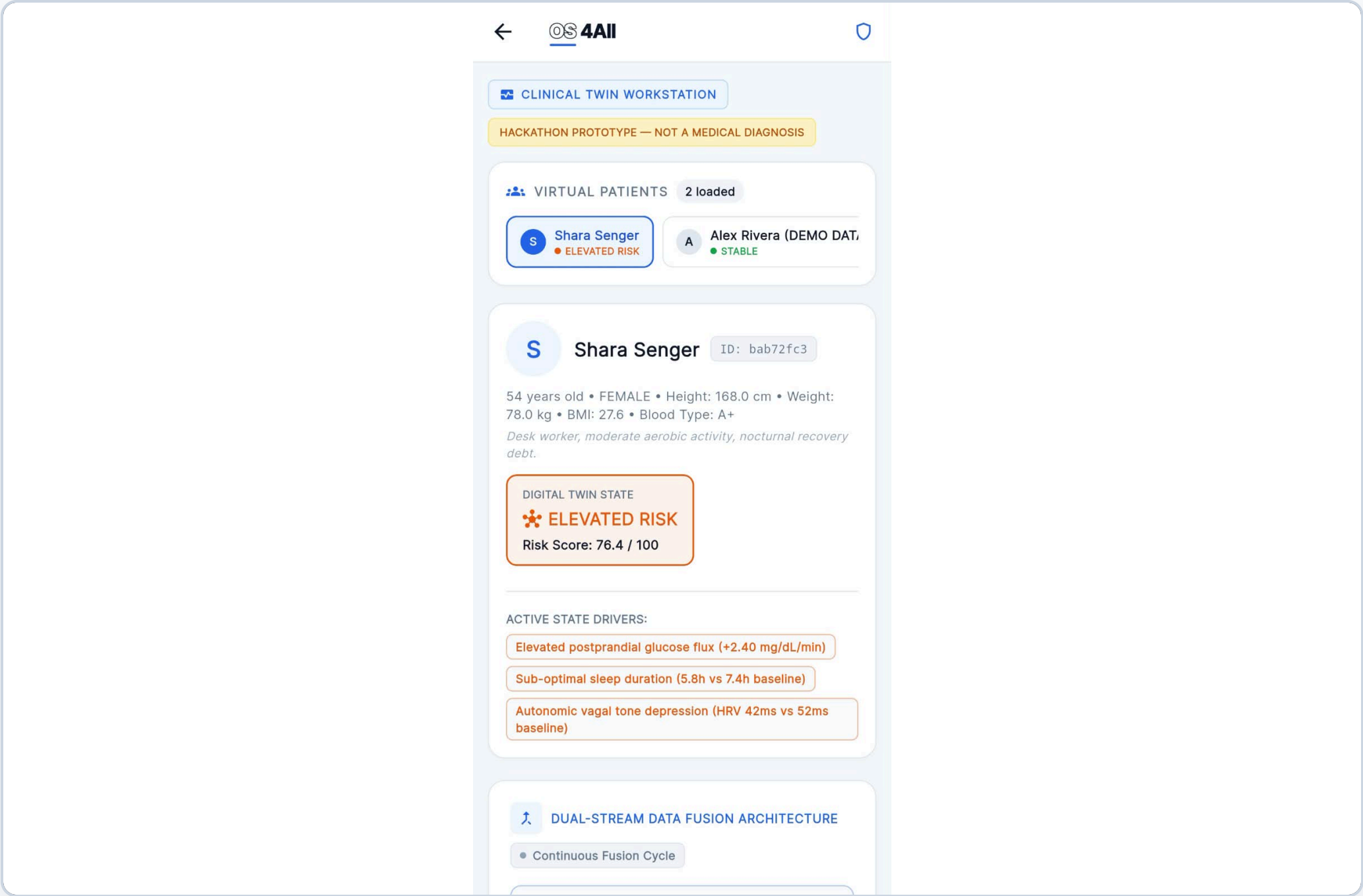
Grounded Clinical Natural-Language Synthesis:

- **Official Google GenAI SDK:** Uses com.google.genai.Client to interact with gemini-3.8-flash.
- **Read-Only Structured Context:** Receives strictly verified snapshot containing patient demographics, baseline deltas, twin state, and feature points.
- **Strict Guardrails:**

- ✗ Gemini NEVER calculates risk scores or trajectories
 - ✗ Gemini NEVER invents biometrics or medical diagnoses
- **Offline Fallback Engine:** Deterministic rule-based templates guarantee 100% functionality without cloud API keys.

Grounded Virtual Patient Q&A: Clinicians in Doctor View can ask questions like "What is driving the predicted spike?". Answers are generated by Gemini strictly using verified active telemetry, ensuring zero hallucinations.

ZERO HALLUCINATIONS



Direct Navigation & Virtual Cohort

Reached via one-click "Doctor View" button on the TWIN screen. Supports switching between distinct synthetic patient profiles: **Alex Rivera**, **Shara Senger**, and **Marcus Vance**.

5 Interactive Simulation Scenarios

Clinicians can inject reproducible perturbation scenarios in real time:
1. Stable • 2. Poor Sleep Drift • 3. High Activity • 4. Glucose Spike • 5. Recovery Walk

Live Trajectory & Feature Contribution

Visualizes the 120-minute forward trajectory curve with milestone cards, paired with an inspectable table showing exact feature points, baseline deltas, and risk directions.

Grounded Virtual Patient Interaction

Interactive Q&A enabling clinicians to ask questions directly to the patient's twin, answered by Gemini using verified runtime telemetry context.

OS4All

DIGITAL TWIN

Your body, modeled.

SHARA SENGERR

Virtual Patient

54 • Female • Type 2 Diabetes

DIGITAL TWIN STATE

ELEVATED RISK

Elevated glycemic surge projected

Risk Score

76.4

/ 100

Current Glucose

162

mg/dL

Velocity

+2.40

mg/dL/min

2-HOUR OUTLOOK

RISING (+170.5 mg/dL)

Deterministic projection: reaches 332.5 mg/dL at T+120m.

Now

162

mg/dL

+30m

205

mg/dL

+60m

254

mg/dL

+90m

298

mg/dL

+120m

332.5

mg/dL

WHAT IS DRIVING THE CHANGE?

Fasting Glucose Baseline Gap

Current 162 mg/dL exceeds personal mean (118 mg/dL) by +37%

High Positive Glucose Rate of Change

Velocity +2.40 mg/dL/min indicates rapid postprandial glucose flux

Reduced Heart Rate Variability

RMSSD 38 ms indicates autonomic stress

HOME

TWIN

LIVE

AI

PROFILE

1. Home Dashboard

Vitals & Health Overview

Your Digital Twin

Active Model

Virtual representation of the patient's current physiological state.

SHARA SENGERR

Virtual Patient

54 • Female • Type 2 Diabetes

CURRENT PHYSIOLOGICAL STATE

ELEVATED RISK

ELEVATED RISK

Postprandial influx accelerating glucose toward upper threshold.

CURRENT PHYSIOLOGY

Current Glucose

162

mg/dL

Sensor reading

Glucose Velocity

+2.40

mg/dL/min

Flux rate of change

Heart Rate

78

BPM

Resting pulse

HRV (RMSSD)

42

ms

Autonomic recovery

Sleep Recovery

5.8

hrs

Nocturnal duration

Daily Activity

2850

steps

Accumulated movement

2-HOUR PROJECTED TRAJECTORY

Deterministic RK4

Target physiological corridor: 70 – 140 mg/dL

HOME

TWIN

LIVE

AI

PROFILE

2. Digital Twin

Twin State & Doctor View

Live Signals

Where is the data coming from? Ingestion streams & device pairings.

DATA SOURCE: PHYSICAL DEVICE

Android Health Connect native records active

Android Health Connect

Connected

Physical on-device biometric store via Android Health Platform.

Blood Glucose

Mapped (mg/dL)

Heart Rate

Continuous (BPM)

Heart Rate Variability

RMSSD (ms)

Steps & Cadence

Daily tally

Sleep Session

Duration & stages

Sync Health Connect Now

Connected • Ready to sync

REAL-TIME SIGNALS

CGM Glucose

162

mg/dL

Continuous stream

Heart Rate

78

BPM

Photoplethysmography

Nocturnal HRV

42

ms

Vagal autonomic index

Sleep Duration

5.8

hrs

Last night recovery

HOME

TWIN

LIVE

AI

PROFILE

3. Live Signals

Continuous IoT Streaming

AI INSIGHTS

Gemini explains what the Digital Twin is seeing.

GEMINI • GROUNDED IN TWIN

Authoritative deterministic model

CURRENT SITUATION

AI-Generated Explanation

Postprandial Metabolic Drift with Elevated Spike Trajectory

The patient exhibits an active upward glucose velocity of +2.40 mg/dL/min following recent dietary intake combined with inadequate nocturnal recovery (5.8 hrs). Under current physiological dynamics, systemic glucose is projected to reach 332.5 mg/dL within 120 minutes without corrective physical exertion.

WHY THIS IS HAPPENING

Fasting Glucose Baseline Gap

Current 162 mg/dL exceeds personal mean (118 mg/dL) by +37%

High Positive Glucose Rate of Change

Velocity +2.40 mg/dL/min indicates rapid postprandial glucose flux

Reduced Heart Rate Variability

RMSSD 38 ms indicates autonomic stress

Sleep Architecture Debt

5.4 hours recorded vs 7.5 hour physiological baseline

Ask Gemini about physiological signals...

HOME

TWIN

LIVE

AI

PROFILE

4. AI Insights

Grounded Interpretations

Profile & Settings

Patient demographics, physiological baselines & system tools.

SHARA SENGERR

Virtual Patient

54 yo • Female • MRN: SEN-9821

Type 2 Diabetes (Synthea FHIR)

Edit Profile & Baselines

PHYSIOLOGICAL BASELINE

Synthea Baseline

Height / Weight

168 cm / 78 kg

Basal Fasting Glucose

118 mg/dL

Target HbA1c

7.0%

Insulin Sensitivity (S_I)

45 mg/dL/U

Carbohydrate Ratio (C_R)

12 g/U

HEALTH PROFILE & NAVIGATION

Clinical Timeline

Longitudinal events & disease progression

Live Demo Mode

Guided 6-step competition pitch presentation

Advanced Intelligence & FHIR

Differential equations, RK4 solver & literature

Application Settings

HOME

TWIN

LIVE

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PROFILE

5. Profile Baselines

45-Day Rolling Envelopes

Clinical Timeline

ALL

SYSTEM

VITALS

LABS

Prototype • Not a medical diagnosis

SYSTEM

Baseline Initialized

Synthea FHIR Cohort Ingested

Diagnosed Type 2 Diabetes (SNOMED 44054006)

VITALS

T - 2 hours

Continuous CGM Stream Attached

5-minute streaming IoT frequency configured

LABS

T - 30 minutes

Glucose Velocity Inflection Detected

Shift from +0.20 to +2.40 mg/dL/min

SYSTEM

T - 5 minutes

Digital Twin State Transitioned

Transitioned to ELEVATED_RISK based on deterministic projection

HOME

TWIN

LIVE

AI

PROFILE

6. Clinical Timeline

Unified Historical EHR

Verified Technical Stack

ARCHITECTURE

- **Backend:** Java 17/21, Spring Boot 3.3.4, Spring Security, Spring Data JPA, Flyway.
- **Frontend:** Flutter 3.x, Dart 3.x, Material 3, fl_chart 0.69.0.
- **Database:** MySQL 8.0 (Production) / H2 In-Memory (Zero-config test).
- **Data Standards:** Synthea™ HL7 FHIR R4 JSON, SNOMED-CT, LOINC, RxNorm.
- **Device Ingestion:** Android Health Connect MethodChannel bridge (API 26+).
- **AI Integration:** Google GenAI Java SDK (gemini-3.8-flash) + Deterministic fallback.
- **Testing:** 87+ Spring Boot unit/integration tests, 24+ Flutter widget tests.

Demonstrated Outcomes

VERIFIED

- **Working Functional Prototype:** Fully functioning end-to-end prototype executing across backend, database, and cross-platform UI.
- **Dual-Stream Data Fusion:** Successfully unites static EHR vulnerability with dynamic 5-minute CGM/IoT biometrics.
- **Personalized Baselines:** 45-day rolling envelopes replace rigid population-wide intervals.
- **Deterministic Prediction:** 2-hour trajectory model projects metabolic curves with milestone cards.
- **Doctor Workstation:** Clinicians inspect cohorts, inject 5 perturbation scenarios, and interact with the twin.
- **Grounded AI Safety:** Strict guardrails ensure zero hallucinated numbers or fake diagnoses.

Future Clinical Roadmap

NEXT STEPS

- **Clinical Prospective Trials:** Partner with clinical academic research centers to evaluate model accuracy against real continuous patient cohorts.
- **Hospital EHR Integration:** Implement live bi-directional SMART-on-FHIR connectors with hospital Epic/Cerner systems.
- **Broader Chronic Conditions:** Expand Digital Twin modeling to cardiovascular disease, heart failure, and renal decline.
- **Regulatory Pathway:** Formal documentation, clinical risk management (ISO 14971), and medical device regulatory submissions (CDSCO / FDA).

Solo Participant Submission: Completed by Kanak Ghawade (Faculty of Eng. & Tech., Wardha / DMIHER).